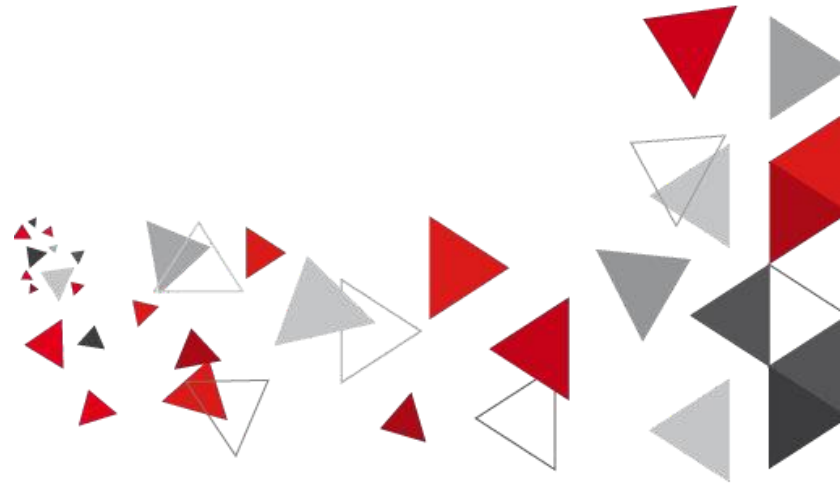




Course 1 - PHP

Introduction to PHP



Plan

- What's PHP?
- Evolution of PHP
- How PHP works?
- Working environment
- Basic element of PHP language
- Basic structure of PHP page
 - Variables
 - Types
 - Arrays
 - Constant
 - Operators
 - Conditional structures
 - Iterative structures
 - Functions

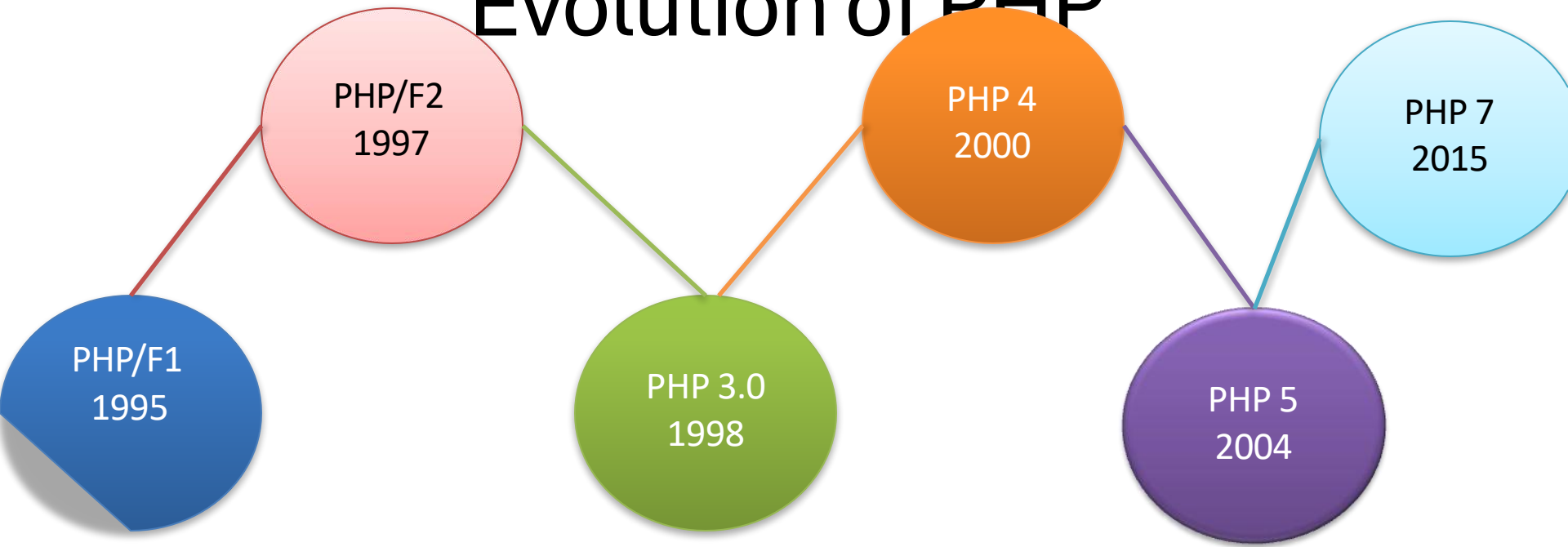


What's PHP?

- A programming language that integrates into HTML pages
- It is an Open Source scripting language
- Designed for developing **dynamic web applications**
- A dynamic, precompiled scripting language
- interpreted server side



Evolution of PHP

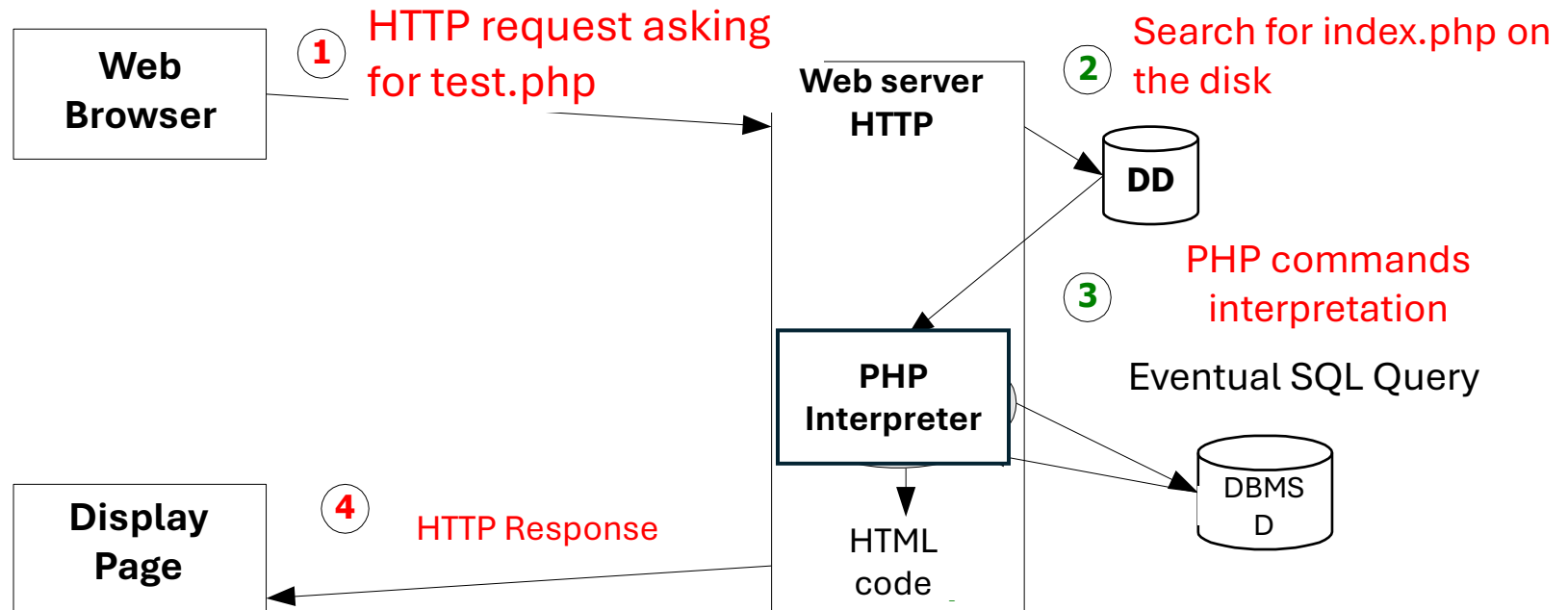




How PHP works ? (1/2)

Client Side

Server Side





XAMPP Control Panel v3.3.0 [Compiled: Apr 6th 2021]



XAMPP Control Panel v3.3.0

 Config
 Netstat
 Shell
 Explorer
 Services
 Help
 Quit

Service	Module	PID(s)	Port(s)	Actions			
☐	Apache	11428 11608	80, 443	Stop	Admin	Config	Logs
☐	MySQL	7280	3306	Stop	Admin	Config	Logs
☐	FileZilla			Start	Admin	Config	Logs
☐	Mercury			Start	Admin	Config	Logs
☐	Tomcat			Start	Admin	Config	Logs

1:08:26 PM [Apache] Status change detected: stopped

1:08:27 PM [mysql] Attempting to stop MySQL app...

1:08:27 PM [mysql] Status change detected: stopped

1:08:37 PM [Apache] Attempting to start Apache app...

1:08:37 PM [Apache] Status change detected: running

1:08:39 PM [mysql] Attempting to start MySQL app...

1:08:39 PM [mysql] Status change detected: running

1:49:12 PM [main] Executing "c:\xampp\"

Working environment:Xampp (2/2)



MySQL is a relational database management system (RDBMS)

Localhost:
Launching local server



Apache HTTP Server (Apache) is an HTTP server created and maintained within the Apache foundation. It is the most popular HTTP server on the World Wide Web.

www directory:
Server base
directory: the root
of the server

phpMyAdmin (PMA) is a management web application for MySQL database management systems made in PHP and distributed under the GNU GPL license



Basic structure of PHP page



PHP instructions are placed between tags:

```
<?php  
// Le texte du script  
?>
```

.php extension page

Comments:

// (a single line) or

/*....*/ (several lines)

```
<?php  
/* Commentaire sur 2  
lignes */  
?>
```

Each statement ends with ;

Variables (1/2)



- A variable **must** start with the \$ sign.
- PHP is case sensitive (upper and lower case), *myvar* and *MyVar* do not mean the same thing.
- You cannot put spaces in a variable name (use underscore _).
- To display the contents of the variable we use the keyword ***echo***

```
<?php
    $myvar = '<p>Hello world!!</p>';
    echo $myvar;
?>
```

Output:
Hello world!

Variables (2/2)



```
<html>
  <head>
    <title> Document PHP </title>
  </head>
  <body>
    <?php
      echo 'Hello world !!!!!';
    ?>
  </body>
</html>
```

Output:
Hello world!



Types (3/6)

There are two types:

- **Scalar types:** these are the types that could be displayed directly by the browser (boolean, integer, decimal, character string).
- **Complex types:** arrays, objects, etc.



Types (1/6)

With PHP we can use the same variable to store and display different types of data.

The variable changes type depending on the content it receives.

- The type of variables is dynamic, light and flexible
- PHP is a weak and dynamic typing language.



Types (2/6)

Example:

```
$myvar=123;           //integer  
$myvar=false;        //Boolean  
$myvar=3.1415;        //float  
$myvar='hello';       //string
```

Scalar types:

Types (4/6)



```
<?php
    $return = '<br/>';
?>

<!DOCTYPE html >
<html>
<head> <title> PHP Document </title> </head>
<body>
    <?php
        $toto=1;
        echo $toto;
        echo $return;
        $toto=1.5;
        echo $toto;
        echo $return;
        $toto='Hello';
        echo $toto;
        echo $return;
    ?>
</body>
</html>
```

Output:

1
1.5
Hello

Types (5/6)



Complex types

```
<?php
    $return = '<br/>';
?>
<!DOCTYPE html >
<html>
<head>
    <title> PHP Document </title>
</head>
<body>
    <?php
        $myvar=array('col1','col2');
        echo $myvar[0];
        echo $return;
        echo $myvar[1];
    ?>
</body>
</html>
```

Output:

col 1
col2



Modification of types

- The desired type can be specified: cast
- `$variable = (nme_of_type) value`

```
$a = (integer) '10';
```

```
$b = (string) 12;
```

```
$c = (float) $b;
```

<code>\$a</code>
<i>integer</i>
10

<code>\$b</code>
<i>string</i>
'12'

<code>\$c</code>
<i>float</i>
12,

Modification of types



```
$nombre1 = 1.5;  
$nombre2 = 12;  
$chaine1 = "10";  
$chaine2 = 'coucou';
```

```
$total =
```

\$total
float
23.5

\$nombre1
float
1.5

\$nombre2
integer
12

\$chaine1
string
'10'

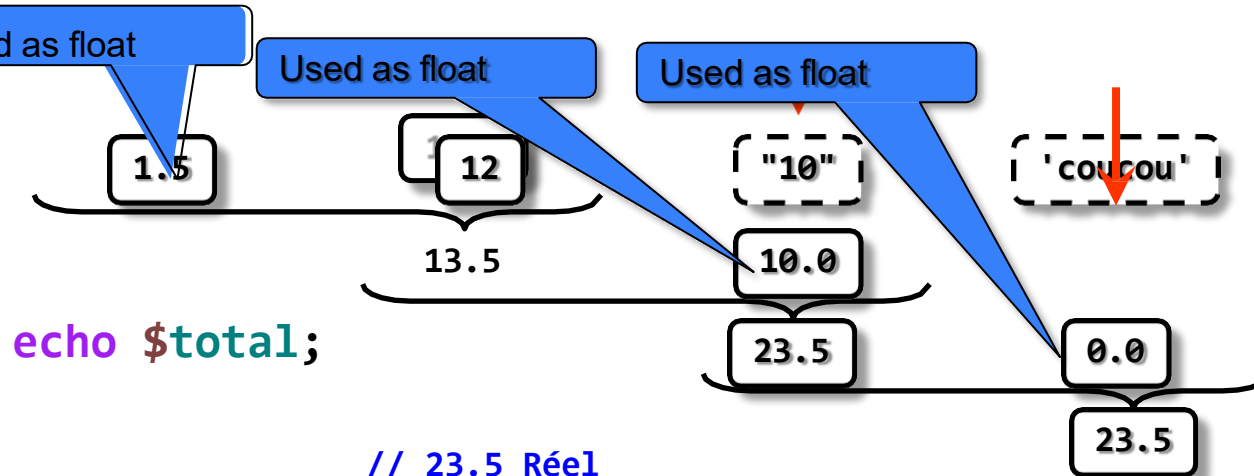
\$chaine2
string
'coucou'

```
as  
$nombre1 + $nombre2 + (float) $chaine1 +(float) $chaine2;
```

Used as float

Used as float

Used as float



```
echo $total;
```

```
// 23.5 Réel
```



Arrays (1/7)

There are two types of tables:

Indexed arrays: Each element of the array is identified by its index (0,1,2,..)

Associative arrays: We associate with each element of the array a key whose value is a string type

Arrays (2/7)



Indexed Arrays: declaration

```
<?php
    $T=array(2,3,6);
    print_r($T);
?>
```

```
<?PHP
    $T=array();
    $T[]=2;
    $T[]=3;
    $T[]=6;
    print_r($T);
?>
```

```
<?PHP
    $T[0]=2;
    $T[1]=3;
    $T[2]=6;
    print_r($T);
?>
```

Output: Array ([0]=>2 [1]=>3 [2]=>6)

Arrays (3/7)



```
<?php
    $T=array( '<p>Hello</p>' , '<p>Bonjour</p>' );
    $content=0;
?>
<!DOCTYPE html >
<html>
    <head>
        <title> PHP document </title>
    </head>
    <body>
        <?php
            echo $T[$content];
        ?>
    </body>
</html>
```

Output:
Hello

Arrays (4/7)



Associative arrays : declaration

```
<?php
    $personne =
array( 'name'=>'Ali',

'prenome'=>'Salah' );
    print_r($personne);
?>
```

```
<?PHP
$personne[ 'name' ]='Ali';
$personne[ 'prenome' ]='Salah'
;
print_r($personne);
?>
```

Output: Array ([name] => Ali [prenome] => Salah
)



Arrays (5/7)

N,B,:

Keys in associative arrays are case sensitive.

`$T['key']` is different from `$T['KEY']`

Strings defining keys must not contain spaces

Arrays (6/7)



```
<?php
    $T=array();
    $T['ch1']='<p>Hello</p>';
    $T['ch2']='<p>Bonjour</p>';
    $content='ch1';
?>
<!DOCTYPE html >
<html>
    <head>
        <title> PHP document </title>
    </head>
    <body>
        <?php
            echo $T[$content];
        ?>
    </body>
</html>
```

output:

Hello



Arrays (7/7)

Predefined functions for arrays:

Function	Explanation
Count(\$T), sizeof(\$T)	Return the number of elements of the array \$T.
In_array(\$var,\$T)	Test if the value of \$var exist in the array \$T.
\$sort(\$T)	alphanumerically sorts array elements and reassigns array indices



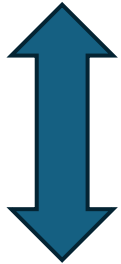
Constants (1/3)

<?php

name

valeur

```
define('MY_CONSTANT', 'Bonjour');
```



Definition of a constant

```
Const MY_CONSTANT = 'Bonjour';
```

name

valeur

Use of a constant:

```
Echo MY_CONSTANT;
```



Constants (2/3)

A constant is a value that can no longer be modified.

```
<?php
    define('AFFICHE', 'Hello');
?>
<!DOCTYPE html>
<html>
<head>
    <title>Document PHP</title>
</head>
<body>
    <?php
        echo AFFICHE;
    ?>
</body>
</html>
```

Output:
Hello



Constants (3/3)

- By convention the name of the constant must be in capital letters.
- To define a constant, we do not use \$.
- The define function returns **TRUE** on success and **FALSE** on failure.

Operators (1/5)



Operators are used to compare, calculate, and combine values.

The main operators are:

Assignment operators.

Concatenation operators: “`.` `>>`”

Arithmetic operators: “`+`, `-`, `*`, `/`”

Comparison operators.

Logical operators.

Operators (2/5)



Arithmetic operators

Example	result
<code>\$a=1; \$b=\$a; echo \$b;</code>	1
<code>\$a=1; echo \$a+=1;</code>	2
<code>\$a=1; echo \$a-=1;</code>	0
<code>\$a=1; echo \$a*=1;</code>	1
<code>\$a=1; echo \$a/=1;</code>	1

Operators (3/5)



Assignment operators

Example	Result
<code>\$a='Hello'; \$b=' ahmed'; echo \$a.= \$b;</code>	Hello ahmed
<code>\$a=1; \$a++; Echo \$a;</code>	2
<code>\$a=1; ++\$a; Echo \$a;</code>	2
<code>\$a=1; echo \$a++; echo '
'; echo \$a; echo '
'; echo ++\$a; echo '
'; echo \$a;</code>	1 2 3 3

Operators (4/5)



Comparison Operators

Operator	Value
<code>==</code>	equal
<code>===</code>	Identical (value and type)
<code><></code>	Not equal
<code>!=</code>	Not equal
<code>!==</code>	Not identical (value or type)
<code><</code>	inferior
<code>></code>	superior
<code><=</code>	Inferior or equal
<code>>=</code>	Superior or equal

Operators (5/6)



```
$a = 12; $b = '12'; $c = 13.0;
```

```
var_dump($a == $b);  
var_dump($a == $c);  
var_dump($c == $b);  
var_dump($a != $b);  
var_dump($a != $c);  
var_dump($c != $b);  
var_dump($a === $b);  
var_dump($a === $c);  
var_dump($c === $b);  
var_dump($a !== $b);  
var_dump($a !== $c);  
var_dump($c !== $b);
```

boolean true
boolean false
boolean false
boolean false
boolean true
boolean true
boolean false
boolean false
boolean true
boolean true
boolean true

\$a
integer
12

\$b
string
'12'

\$c
float
13

Operators (6/6)



Logical Operators

Opérateur	Valeur
\$a and \$b	True if \$a is true and \$b is true
\$a or \$b	True if \$a is true or \$b is true
\$a xor \$b	True if \$a is or \$b is true but not the two at the same time
! \$a	True if \$a is not true
\$a && \$b	True if \$a is true et \$b is true
\$a \$b	True if \$a is true or \$b are true

Conditional statement (1/4)



```
if (condition) {  
    /* Block of instructions executed  
       if the condition is true*/  
} else {  
    /* Block of instructions executed  
       if the condition is false*/  
}
```

Optional

Conditional statement (2/4)



if else

```
<?php
define('AFFICHE','ch2');
$ch1='<p>Hello</p>';
$ch2='<p>Bonjour</p>';
$ch3='<p>Salut</p>';
?>
<!DOCTYPE html>
<html>
<head>
    <title>Document PHP</title>
</head>
<body>
    <?php
    if (AFFICHE=='ch1')
    {
        echo $ch1;
    }
    else if (AFFICHE=='ch2')
    {
        echo $ch2;
    }
    else
    {
        echo $ch3;
    }
    ?>
</body>
</html>
```

Output:

Bonjour

Conditional statement (3/4)



switch (EXP1)

```
<?php
    define('AFFICHE','ch3');
    $ch1='<p>Hello</p>';
    $ch2='<p>Bonjour</p>';
    $ch3='<p>Salut</p>';
    ??
<!DOCTYPE html>
<html>
<head>
    <title>Document PHP</title>
</head>
<body>
    <?php
    switch (AFFICHE)
    {
        case 'ch1':
            echo $ch1;
            break;
        case 'ch2':
            echo $ch2;
            break;
        case 'ch3':
            echo $ch3;
            break;
        default:
            echo 'votre chaine n\'existe pas';
    }
    ??
</body>
</html>
```

Output:
Salut

Conditional statement (4/4)



switch (EXP2)

```
<?php
    define('AFFICHE','ch3');
    $ch1='<p>Hello</p>';
    $ch2='<p>Bonjour</p>';
    $ch3='<p>Salut</p>';
    ??
<!DOCTYPE html>
<html>
<head>
    <title>Document PHP</title>
</head>
<body>
    <?php
        switch(AFFICHE):
            case'ch1':
                echo $ch1;
                break;
            case'ch2':
                echo $ch2;
                break;
            case'ch3':
                echo $ch3;
                break;
            default:
                echo 'votre chaine n\'existe pas';
        endswitch;
    ??
</body>
</html>
```

Output:
Salut

Iterative Statement (1/4)



for

```
<?php
    $retour='<br/>';
?>
<!DOCTYPE html>
<html>
<head>
    <title>Document PHP</title>
</head>
<body>
    <?php
        for($i=0;$i<=3;$i++)
        {
            echo $i.$retour;
        }
    ?>
</body>
</html>
```

Output:

0
1
2
3

Iterative statement (2/4)



while

```
<?php
    $retour='<br/>';
?>
<!DOCTYPE html>
<html>
<head>
    <title>Document PHP</title>
</head>
<body>
    <?php
        $i=0;
        while ($i<=3)
        {
            echo $i.$retour;
            $i++;
        }
    ?>
</body>
</html>
```

Output:

0
1
2
3



Iterative statement (3/4)

```
<?php
    $retour='<br/>';
?>
<!DOCTYPE html>
<html>
<head>
    <title>Document PHP</title>
</head>
<body>
    <?php
        $i=0;
        do
        {
            echo $i.$retour;
            $i++;
        }
        while ($i<=3);
    ?>
</body>
</html>
```

Output:

0
1
2
3

Iterative statement (4/4)



foreach

```
<?php
    $retour='<br/>';
?>
<!DOCTYPE html>
<html>
<head>
    <title>Document PHP</title>
</head>
<body>
    <?php
        $i=array(1,2,3,4);
        foreach($i as $valeur)
        {
            echo $valeur.$retour;
        }
    ?>
</body>
</html>
```

Output:

1
2
3
4



Functions definition

- Function
- `function_name() {`
 - `instruction1; instruction2; ...`
- `}`
- Example:
- `function calculSomme($a, $b)`
- `{`
 - `return $a+$b;`
- `}`

References



- ▶ <http://www.php.net>
- ▶ <http://www.phpinfo.net>
- ▶ <http://www.phpfrance.com>
- ▶ <http://www.developpez.com/php/>

Pour obtenir en ligne toute la documentation officielle (en français) sur une commande PHP :

http://fr.php.net/nom_de_la_commande

Exemple : <http://fr.php.net/echo>